Project plan

Keuzewijzer

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Version Control

Version	Date	Description
0.1	09-09-2024	Initial Project Plan
0.2	10-09-2024	Ready for feedback
0.3	18-12-2024	Language changes
1.0	25-12-2024	Document ready

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1. Introduction

In today's educational environment, students encounter a wide variety of course options and career paths, which can make selecting appropriate courses and planning their academic future complex. At Windesheim, the process of creating an academic plan aligned with career goals can be challenging due to the high level of flexibility and limited guidance in traditional systems. These methods may result in misaligned course selections and an academic plan that may not fully support long-term career objectives.

This document outlines the problem at Windesheim, the scope within which the project will be carried out, the products to be delivered, the methodologies to be used, and the potential risks that may arise during the project. In summary, this document serves as an agreement for the project.

2. Project definition

The Keuzewijzer project aims to improve the course selection process for Windesheim students by providing an interface that allows students to select their academic years and semesters, while presenting multiple learning paths based on predefined criteria.

2.1. Scope

The Keuzewijzer project aims to develop a web-based tool that empowers students to select their academic year, semester, and available learning paths. The system will provide personalized guidance based on students' academic preferences and career interests, ensuring that their choices align with Windesheim's curriculum. This project incorporates the design of the user interface and collaboration with academic advisors to ensure the accuracy and alignment of course pathways with institutional requirements.

All semester-related information will be accessible on Sharenet. However, updating this information is not within our scope.

2.2. Goals

- Developing a streamlined course and semester selection process for students.
- Offering students various learning paths aligned with their career goals.
- Minimizing complexity and providing clear guidance in the academic planning process.
- Simplifying course management to improve the efficiency of the student planning process.

2.3. Needs

The Keuzewijzer project addresses the challenges students encounter when managing course selection and career planning in traditional systems, which often involve a high level of flexibility and lack of clear guidance. This can result in students selecting courses that lead to unstructured academic paths. The Keuzewijzer project aims to address this by developing a tool that combines flexibility with structured guidance, enabling students to make informed academic decisions that align with their career goals.

2.4. Assumptions

- **Student Awareness:** Students are expected to have a basic understanding of how to use digital tools and navigate online platforms.
- **Accurate Course Data:** It is assumed that the course and program data provided by Windesheim is accurate and up to date.
- **University Support:** Windesheim will support the project by providing access to necessary resources like having access to servers, adequate time for research, development, design and testing.
- **Technical Infrastructure:** The university's IT infrastructure will support the development and deployment of the Keuzewijzer.

2.5. Dependencies

- **Data Integration:** The project depends on the accurate integration of Windesheim's course databases, program structures and academic rules.
- **IT Infrastructure:** The system's performance depends on Windesheim's existing IT infrastructure, such as server availability and network capabilities.
- **Collaboration with University:** Continuous collaboration with academic advisors is necessary to ensure that learning paths are accurately mapped and aligned with academic programs.

2.6. Terms of Reference

- **Stakeholders:** The key stakeholders include Windesheim University administration, faculty members, IT staff, and students. Each group will play a critical role in the effective implementation and proper usage of the platform.
- **Project Timeline:** The project is expected to be completed within a specified timeframe (20 weeks), including design, development, testing and deployment phases.
- **Performance Metrics:** Success will be measured by user satisfaction, student adoption rates, system reliability and course selection efficiency improvements.
- **Budget:** The project will operate within a predefined budget that covers development costs, IT resources and ongoing maintenance.
- **Maintenance and Support:** Post-launch, the Keuzewijzer system will require ongoing support and updates to ensure that it continues to meet the needs of the students and faculty as course offerings and academic rules evolve.

3. Projectmanagement organization

This chapter provides information about the stakeholders within this project. This means we will show and explain how much interest they have for this project and the amount of influence they can exert on the final product. Furthermore, we will be talking about how the development team is structured and which roles everyone has.

Stakeholder analysis

Within this project, several stakeholders are involved. The involvement of the stakeholders is made clear through the power-interest matrix below.



Figure 1Power-Interest Matrix

Stakeholder	Involvement
Teun Lucassen	Is the product owner of this project.
Arnoud Kuijt	Learn route advisor
Student Coaches	Student coaches will be using the tool to help students choose a path for the certificate they are going to receive when finishing their degree. That is also why student coaches will be one of the most frequent users of this tool.

HBO-ICT student	HBO-ICT students will be using to see what path they can follow for the degree they want to get. The students can use this tool whenever they want to look up and see if a course fits their expectations and will be the most frequent users of this tool.
Companies	Windesheim shows the Keuzewijzer to companies so that companies are able to discern what level the students have acquired.
Parents	Parents can see what their children will be learning on Windesheim via the use of this tool.

3.1. Team Organization

Within the development team, several roles have been assigned. These roles ensure the distribution of general tasks, such as communication, maintaining the schedule, etc. Additionally, this approach always ensures that there is a person with final responsibility. It provides clear structure within the team. The following roles have been assigned:

Role	Description
Team Manager	The team manager coordinates team activities, sets goals, and manages resources. They ensure the team meets its objectives, provide support and guidance, and address any issues that arise. They also facilitate communication within the team and with stakeholders.
Scrum Master	The Scrum Master helps the development team follow Scrum practices, removes obstacles, facilitates meetings, and ensures effective communication between the team and stakeholders.
Team Member	Responsible for realizing the given tasks and providing the increments which follow the Definition of Done (DoD).

The diagram below shows the current role distribution within the team.

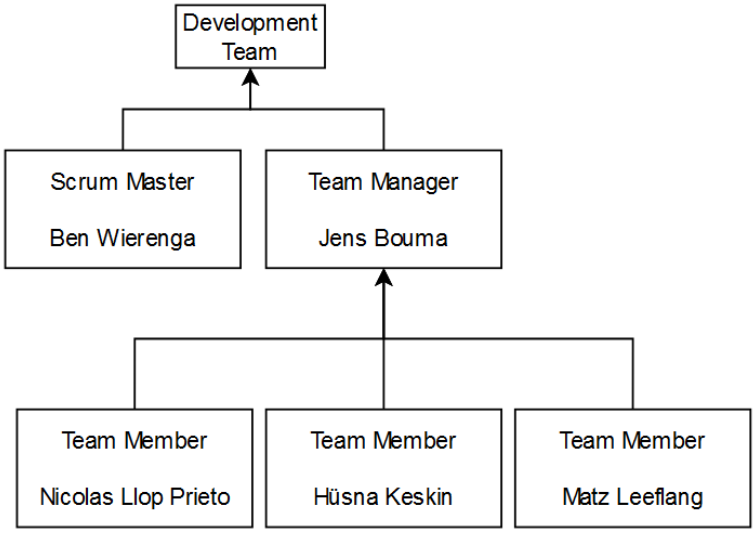


Figure 2 Team Structure

4. Products

This chapter provides a detailed description of all the products that will be created during the project. Each product is explained with a description of the methodology, techniques used, justification for these techniques, quality requirements, and the way this quality will be ensured in the acceptance criteria.

4.1. Product Breakdown Structure

In the Product Breakdown Structure (PBS), it is visible which products are going to be delivered and under which competency they fall.

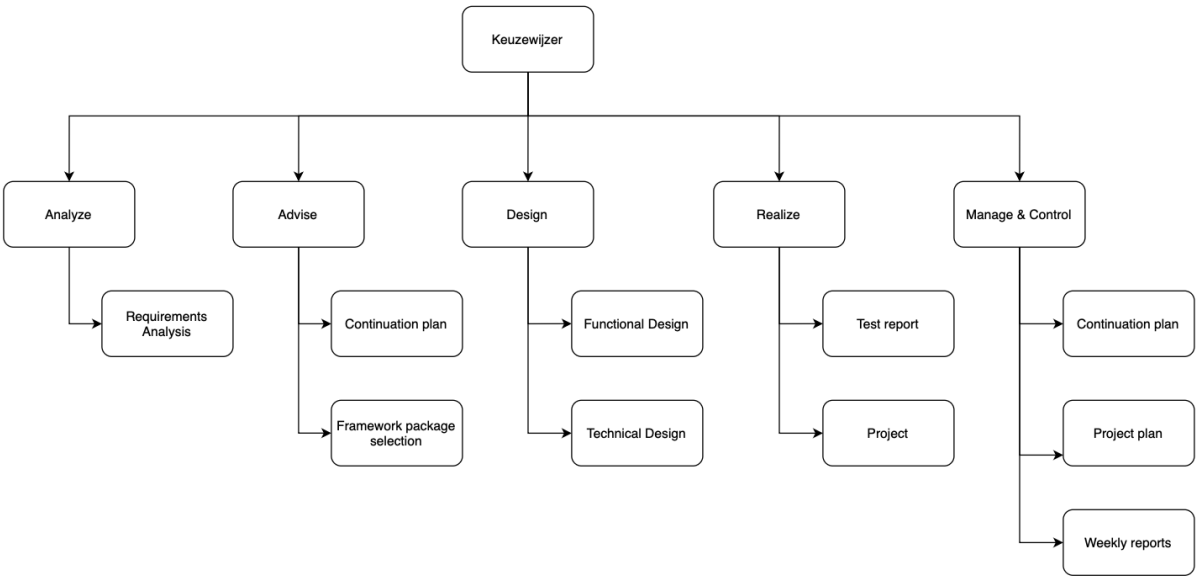


Figure 3 Product decomposition structure

4.2. Product descriptions

In this paragraph, the products are described and explained individually. For each product, you can read what is described and the purpose for which the product is delivered. Each product also has a list of acceptance criteria that it must meet. These criteria ensure the quality of the product.

Project Plan

Product	Project Plan
Description	The Project Plan contains all the necessary information to manage this project efficiently. The Project Plan includes the goal, products, activities, methods and planning for this project.
Activities	The products will be briefly described. Each product will also be appointed activities which are needed to make the product. Additionally, each product will be assigned a list of acceptance criteria that it must meet.
Acceptance Criteria	- Includes a clear description of the project. - Includes a description of the client. - Project boundaries are specified. - Includes description of all the projects that will be realized. - Activities are recorded in the planning. - Management tools are described.

Requirement analysis

Product	Requirement analysis
Description	To get a clear understanding of the requirements for the product, a requirements analysis is conducted. The requirements analysis includes a list of criteria that the product must meet. These requirements are then used to formulate the final User Stories.
Activities	Interview with the product owner about the functional and non-functional requirements which are needed for the Keuzewijzer.
Acceptance Criteria	- The questions and answers from the interview are noted.

Functional Design

Product	Functional Design
Description	In the functional design the acquired in the Requirement analysis are worked out in SMART user stories. These user stories are then worked out and further explained with Use Cases, Use Case diagram and domain model.
Activities	Make user stories from the acquired requirements. These user stories will be used to create Use Cases and a Use Case diagram
Acceptance Criteria	- All User Stories have SMART acceptance criteria. - Use Cases and Use Case diagram created. - There are screen mockups.

Technical Design

Product	Technical Design
Description	The Technical Design has every technical aspect of the project. All information that the developers may need for further development are located in this document. The technical information within this document includes used techniques, infrastructure and a C4-model.
Activities	The used coding conventions and techniques are established. Additionally, descriptions about the infrastructure of this project are made. furthermore, there will be a C4-model which further explains the project.
Acceptance Criteria	- C4-model which each level of the model explained.

Test Report

Product	Test Report
Description	The Test Report consists of results of tests for the functionality of the product.
Activities	Follow the tests and type down the results.
Acceptance Criteria	- All tests have a result. - Tests must be performed by someone that did not realize the functionality.

Keuzewijzer

Product	Keuzewijzer
Description	The Keuzewijzer is the final product that will be realized. This Keuzewijzer is a tool that helps students of the HBO-ICT Windesheim making a choice on which semesters they want to follow.
Activities	Making a front- and backend for the Keuzewijzer. Make it modular for future improvements.
Acceptance Criteria	- Redirect to course information on Sharenet. - Able to add delays in learning path. - Courses which have prerequisites can't be chosen if those are not fulfilled.

Framework Package Selection

Product	Framework Package Selection
Description	The Framework Package Selection is a document in which we research which framework best fits our needs and uses. In this document, we will advise the client on the suggested frameworks, after which the client can choose one to be used.
Activities	Make a longlist of frameworks and code language that is used with the frameworks. Make a list of requirements and criteria. Make a short list of the frameworks which passes these requirements and criteria. Give an advice based on the short list.
Acceptance Criteria	- Requirements and criteria must have meaning towards filtering the frameworks. - The longlist should have at least had multiple frameworks per code language.

Weekly Report

Product	Weekly Report
Description	Weekly report serves as a communication document to let the client know how much progression we have made that week and what problems have arisen.
Activities	Work out what we have done for each competence that week and type down if there were any problems involved with the project on which the client is needed.
Acceptance Criteria	- Every field is filled in.

Continuation plan

Product	Continuation plan
Description	The Continuation plan is the document with all the necessary information to get started for the team that takes over this project.
Activities	Outline the tools and platforms being used, with instructions on access and usage. Detail the project roadmap and next steps for the new team.
Acceptance Criteria	- The document is clear, complete, and ready for the next team to start working. - All key information is included, and no critical details are missing.

4.3. Management Products

During this project, the Scrum framework is used. Scrum is a method of Agile working that enables the creation of products in a flexible way. As a result, several management products from Scrum are utilized.

- Backlog refinement
- Sprint planning
- Sprint retrospective
- Burndown chart

4.4. Product Flow Diagram

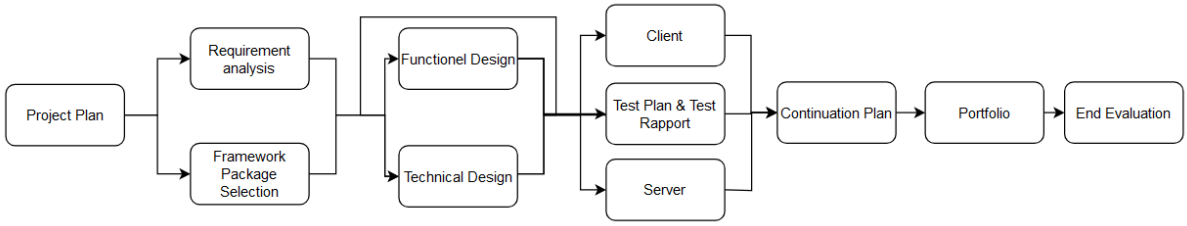


Figure 4 Product flow diagram

This paragraph contains a diagram of the product flow that we will follow for the project. The products are described in chronological order in the diagram.

There are also products that are not displayed in the diagram. These are the products that are not dependent on the completion of other products and are recursive. For example, the Scrum board and time tracking forms are updated daily, and products like retrospectives and burndown charts are updated per sprint.

5. Project Approach

The chosen methodology for the development of this project is Scrum, an agile framework that aligns well with the dynamic and evolving nature of our web development project. Scrum provides the flexibility and adaptability necessary to accommodate the ever-changing requirements and scope of our product.

Given that the product is not fully defined during its early stages, the scope and specifications will require continuous reassessment as new requirements or challenges arise. This iterative process allows us to respond swiftly to feedback and evolving stakeholder needs, ensuring that the product evolves in the right direction. The project plan remains open for revision, enabling us to incorporate new insights and refinements as we make progress toward fulfilling the identified requirements of our various stakeholders.

Our development process will be structured into sprints, which are time-boxed iterations where specific objectives are targeted. During each sprint, we will focus on completing these objectives within a predetermined timeframe, after which we will conduct a review to assess progress, evaluate our work, and determine if any adjustments to our approach or strategy are needed. Initially, these sprints will span four weeks during the early development phase, allowing us to lay a strong foundation. As we move forward and the project matures, we will reduce the sprint duration to two weeks to enable closer monitoring of progress and faster adaptation to any emerging changes or challenges.

By adopting the Scrum methodology, we aim to foster better collaboration, maintain high flexibility, and deliver continuous value. This approach not only supports the team's efforts to meet project milestones efficiently but also ensures that we can respond to new developments with agility, keeping our stakeholders evolving needs at the forefront of the development process.

6. Methods

6.1 Research methods

To create a comprehensive Keuzewijzer for HBO ICT, multiple research methods will be used to gather a wide range of insights and perspectives. These methods include surveys, interviews, a literature study, exploring user requirements and a stakeholder analysis, each contributing to a better understanding of the product goal.

6.1.1 Surveys

Surveys are conducted with teachers, Study Career Counselors, and other key stakeholders to get better insights of their experiences with the previous Keuzewijzer and the expectations regarding the HBO-ICT study. This method provides a broad sample of opinions.

6.1.2 Literature

A literature study is undertaken to explore existing research and reports about the previous Keuzewijzer or another research that is related to a Keuzewijzer.

6.1.3 Interviews

Interviews are done to get information from the different stakeholders on what they think of the Keuzewijzer. This method allows for asking a lot of questions directly to a stakeholder for direct feedback.

6.1.4 Explore user requirements

This helps with getting a detailed view of how users will be using the solution and what their requirements are. This is done by asking the different users how they will interact with the system.

6.1.5 Stakeholder analysis

A stakeholder analysis is done by thinking about the stakeholders who are important to your project and analyzing the importance and interests of each stakeholder. This is important as it helps identifying the stakeholders and ensuring their needs are considered.

6.2 Testing methods

To make sure such a big project goes well multiple testing methods will be used to have a good quality check.

6.2.1 Unit testing

Unit testing can help with finding bugs as early as possible and ensures your code keeps running after a change. This is done by defining one or more test for each ‘atomic part’ of the code like a method or a function. This testing should be done in isolation.

6.2.2 Non-functional testing

Non-functional testing refers to the testing of a software system’s non-functional aspects such as performance, usability, reliability, scalability, and security, rather than the specific behaviors or features of the system. It focuses on how well the system works rather than what it does.

6.2.3 Peer reviewing

Peer review is a process in which work or code is evaluated by others who are knowledgeable in the same field to verify quality, accuracy, and adherence to standards. For this project this will be done by talking with other groups in the semester which will help with getting input from other people with similar knowledge to us.

6.2.4 Code reviewing

Code review is the process where developers evaluate each other’s code to ensure that it meets specific quality standards, adheres to best practices, and is free from errors. This will be done in this project by other group members who did not work on the code that is being evaluated to ensure quality.

6.3 Prototyping

Prototyping is the process of creating an early model or mock-up of a product, system, or feature to explore and validate design ideas before full development. It serves as a preliminary version of the final product and allows stakeholders to interact with a basic version of the design, providing feedback on its usability, functionality, and overall flow.

This is done in this project by making a prototype of the Keuzewijzer which only includes the paths to give the stakeholders a view of what it looks like and how it will work.

6.4 Communication

Communication within the group takes place via WhatsApp and Teams. WhatsApp is used for quick and brief messages, while Teams is used for more document related messages. When it comes to contact with stakeholders, this is done via email.

6.5 Document Management

Documents are hosted online in Teams in Word format, allowing each team member to access them. When documents are delivered, they are submitted in PDF format.

6.6 Style

We use the same template for all the products we need to deliver. This template is stored in the Teams channel of our project.

6.7 Tools

We use the following tools:

- **Word**: Documents are drafted in this.
- **Trello**: Tasks are divided and planned in a clear manner.
- **Figma**: Figma will be used to create more advanced designs for the Keuzewijzer.
- **GitHub:** GitHub will be used for our repository.
- **Draw.io:** drawi.io is one of the tools we will be using to create diagrams and other figures for our documents.
- **Lucidchart:** Lucidchart is another tool that we will be using for creating diagrams and figures if draw.io doesn't full fill our needs.

7. Risks

During the project, certain risks may arise within the team or around the project itself. It is important to analyze these potential risks in advance and consider how we will manage them. For each risk, we evaluated the likelihood of the risk occurring and its potential impact. This was primarily based on assessments of past projects and logical reasoning.

7.1. Project risks

Risk	Probability	Impact	Measure	Result
The project may fall outside the original scope.	●●●○○	●●●○○	We use Scrum, which allows us to work agile, monitor progress closely, and adjust where necessary. Anything that is not completed on time will be included in a recommendation report.	Probability: ●○○○○ Impact: ●●○○○
Something unpredictable goes wrong	●●●●●	?	Rules about who to contact and first steps on what to do are set in place.	Probability: ●●●●● Impact: ?
The project exceeds the agreed-upon budget.	●●○○○	●●●○○	If this turns out to be the case, we will inform the stakeholders as soon as possible to discuss how to proceed.	Probability: ●●○○○ Impact: ●●○○○
Not everything can be implemented on time.	●●○○○	●●●○○	Scrum is used during the project to keep track of our progress. If something is not completed on time, we will inform the stakeholders as soon as possible.	Probability ●○○○○ Impact: ●●○○○
A team member has decided to suspend their studies/semester and has removed themselves from the project group.	●●○○○	●●●○○	We monitor each other to keep everyone's motivation high. If issues arise, we will inform the relevant stakeholders as soon as possible.	Probability: ●●○○○ Impact: ●●○○○
One or more team members are absent or sick for an extended period during the project.	●●○○○	●●●○○	If this is the case, we will inform the stakeholders as soon as possible and adjust our scope as needed in consultation with them.	Probability: ●●○○○ Impact: ●●○○○

The produced items do not meet the agreed-upon criteria.	●●○○○	●●●○○	A completed product is reviewed by at least one person other than the one who created it.	Probability ●○○○○ Impact: ●●○○○
Only one person has access to a file	●●○○○	●●●○○	The team works in places that everyone can access like Teams	Probability ●○○○○ Impact: ●●○○○
A file is lost	●●○○○	●●●○○	A backup will be created and updated	Probability ●○○○○ Impact: ●●○○○
Stakeholders not available	●●●○○	●●●○○	Meetings will be scheduled far ahead of time	Probability ●○○○○ Impact: ●●○○○
Issues with technology or tools that impede project progress or quality.	●●○○○	●●●○○	Conduct thorough testing and quality assurance. Maintain and update technical documentation	Probability ●○○○○ Impact: ●●○○○
Lack of access to required resources, including personnel, equipment, or materials.	●●○○○	●●●○○	Ensure resource planning and allocation are discussed ahead of time	Probability ●○○○○ Impact: ●●○○○
Team members lack the necessary skills or expertise to complete their tasks effectively.	●●○○○	●●●○○	Assess team skills regularly and address gaps promptly	Probability ●○○○○ Impact: ●●○○○
Ineffective communication within the team or with stakeholders, leading to misunderstandings and misaligned goals	●●○○○	●●●○○	Establish clear communication channels and protocols. Hold regular team meetings and status updates. Use collaboration tools to ensure information is accessible.	Probability ●○○○○ Impact: ●●○○○
Conflicting interests or priorities among stakeholders that impact project decisions and progress.	●●○○○	●●●○○	Engage stakeholders early and often. Use structured decision-making processes to address disagreements.	Probability ●○○○○ Impact: ●●○○○

New or changing regulations that affect project requirements, timelines, or costs.	●●●○○	●●●○○	Stay informed about relevant regulations and industry standards	Probability ●○○○○ Impact: ●●○○○
Interpersonal conflicts within the team that disrupt collaboration and productivity	●●○○○	●●●○○	Foster a positive team culture and address conflicts promptly and professionally. Implement team-building activities to improve cohesion.	Probability ●○○○○ Impact: ●●○○○
Unauthorized access or loss of sensitive information affecting project integrity and stakeholder trust	●●○○○	●●●○○	Implement strong security protocols and data protection measures.	Probability ●○○○○ Impact: ●●●○○
Ambiguities or misunderstandings in project requirements leading to misalignment and rework.	●●●○○	●●●○○	Engage stakeholders to clarify and document requirements. Use iterative development and feedback loops to refine requirements.	Probability ●○○○○ Impact: ●●○○○
Team or stakeholder resistance to changes in project scope, processes, or technology.	●●○○○	●●●○○	Communicate the benefits and rationale for changes clearly. Involve stakeholders in the change process and seek their input.	Probability ●○○○○ Impact: ●●○○○

8. Planning

8.1. Main phases

For the project planning, a GANTT chart is used to visualize the phases of Keuzewijzer management, spanning from project analysis to deployment.

Products / Weeks	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16	Week 17	Week 18	Week 19	Week 20
Analysing																				
Designing																				
Realising																				
Testing																				
Deploying																				

Figure 5GANTT-chart planning

8.1.1. Analyzing

During the analyzing phase the focus is on understanding the project requirements and goals.

8.1.2. Designing

During the designing phase the focus is on designing the conceptual and technical aspects of the project are mapped out.

8.1.2. Realizing

During the realizing phase the focus is on the actual development of the project.

8.1.2. Testing

During the testing phase together with the realizing phase the focus is on testing the built features from the realizing phase. This phase will ensure the features that are built works as expected.

8.1.2. Deployment

During the deployment phase the focus is on deploying the built project so the project can be launched and used by the stakeholders.

8.2. Milestones

The project will have several key milestones, each marking the completion of a significant phase or deliverable. These milestones include:

- **Completion of Analysis**: Finalization of the requirements and project scope.
- **Design Sign-off**: Approval of the conceptual and technical design.
- **Feature Complete**: Development phase where all primary features are implemented.
- **Testing Sign-off**: Validation that the system has passed all testing stages.
- **Project Deployment**: Launch of the final product for the stakeholders.

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HBO-i. (2018). *ICT Research Methods*. Opgehaald van Methods Pack for Research in ICT. ICT Research Methods.: https://ictresearchmethods.nl/